

A Variance-Minimizing Birth-Death Process for Two Evolving Sock Clusters

Abstract

This report formalizes a stochastic sock-selection process consisting of two initially pure clusters, a White cluster and a Black cluster. At every iteration, participants draw four socks, select the closest pair, update their values, and decide whether unused socks should be retained or discarded. Additional mechanisms include age-dependent removal at cluster boundaries and population replenishment. The objective is to maintain minimal within-cluster variance while preserving clear separation between the two clusters.

1 Problem Definition

Consider a box containing N socks. Initially,

$$\frac{N}{2}$$

socks belong to the White cluster and have value

$$255,$$

while the remaining

$$\frac{N}{2}$$

socks belong to the Black cluster and have value

$$0.$$

At each iteration:

1. A participant randomly draws four socks.
2. The pair having the shortest value distance is selected.
3. The selected socks are updated and returned.
4. The remaining two socks may be returned or discarded.

2 Sock Value Evolution

2.1 White Cluster

For a white sock,

$$x_W \leftarrow \max(x_W - 2, 160).$$

Thus a white sock gradually evolves as

$$255 \rightarrow 253 \rightarrow 251 \rightarrow \cdots \rightarrow 160.$$

2.2 Black Cluster

For a black sock,

$$x_B \leftarrow \min(x_B + 1, 60).$$

Thus a black sock evolves as

$$0 \rightarrow 1 \rightarrow 2 \rightarrow \dots \rightarrow 60.$$

3 Closest-Pair Selection

When four socks are sampled, the pair with minimum value distance is selected. Since White and Black clusters remain well separated,

$$|x_W - x_B| \gg |x_W^{(1)} - x_W^{(2)}|$$

and

$$|x_W - x_B| \gg |x_B^{(1)} - x_B^{(2)}|.$$

Consequently, the closest pair is almost always drawn from the same cluster. This naturally preserves cluster identity throughout the process.

4 Variance Objective

Let

$$\mu_W$$

and

$$\mu_B$$

denote the means of the White and Black clusters. The cluster variances are

$$V_W = \frac{1}{N_W} \sum_{i=1}^{N_W} (x_i - \mu_W)^2,$$

and

$$V_B = \frac{1}{N_B} \sum_{i=1}^{N_B} (x_i - \mu_B)^2.$$

The optimization objective is

$$\min(V_W + V_B).$$

Maintaining low variance corresponds to keeping both clusters as compact as possible.

5 Unused Sock Management

The principal source of variance comes from outlier socks. For each unused sock, define its deviation score

$$s_i = |x_i - \mu_c|,$$

where μ_c denotes the mean of its corresponding cluster. The variance-minimizing rule is:

1. Compute the deviation score of each unused sock.

2. Retain socks that are close to their cluster center.
3. Preferentially discard socks with the largest deviation.

This strategy continuously removes variance-producing outliers.

6 Boundary Aging Mechanism

A sock becomes saturated when it reaches its cluster boundary.

6.1 White Boundary

A white sock reaches

$$x_W = 160.$$

6.2 Black Boundary

A black sock reaches

$$x_B = 60.$$

After reaching the boundary, the value remains fixed. Let

$$k$$

denote the number of selections experienced after reaching the boundary.

6.3 Final Aging Rule

To preserve the original exponential-aging intuition while ensuring increasing removal pressure, the discard probability is

$$P_k = 1 - p^k, \quad 0 < p < 1.$$

Examples:

$$P_1 = 1 - p,$$

$$P_2 = 1 - p^2,$$

$$P_3 = 1 - p^3.$$

The probability increases monotonically with age:

$$P_{k+1} > P_k.$$

Furthermore,

$$\lim_{k \rightarrow \infty} P_k = 1.$$

Thus very old saturated socks are eventually removed with near certainty.

7 Survival Probability

The survival probability after k post-boundary selections is

$$S(k) = \prod_{i=1}^k (1 - P_i).$$

Since

$$1 - P_i = p^i,$$

we obtain

$$S(k) = \prod_{i=1}^k p^i = p^{\frac{k(k+1)}{2}}.$$

Therefore survival decays faster than a standard geometric process.

8 Population Replenishment

Whenever six socks have been discarded,

$$D = 6,$$

six new socks are inserted. To preserve symmetry:

$$3$$

new White socks with value

$$255$$

are inserted, and

$$3$$

new Black socks with value

$$0$$

are inserted. This creates a balanced birth-death process.

9 Recommended Online Control Strategy

The complete algorithm is:

1. Draw four socks uniformly at random.
2. Select the closest pair.
3. Update the selected pair.
4. Return the selected pair to the population.
5. Compute deviation scores for the unused socks.
6. Retain representative socks.
7. Discard strong outliers.

8. Apply boundary-aging removal using

$$P_k = 1 - p^k.$$

9. Whenever six socks have been removed, insert

3

new White socks at 255 and

3

new Black socks at 0.

10 Expected Behavior

The proposed system exhibits the following properties:

- White and Black clusters remain well separated.
- Outliers are progressively removed.
- Old saturated socks are eliminated through increasing aging pressure.
- Newly injected socks continually refresh cluster purity.
- The process converges to a dynamic steady state with bounded variance.

11 Future Research Directions

Several theoretical extensions are possible:

1. Markov chain analysis of the population dynamics.
2. Mean-field approximation of cluster variance.
3. Reinforcement learning for optimal discard decisions.
4. Analytical derivation of steady-state distributions.
5. Multi-cluster generalizations.

12 Reinforcement Learning for Optimal Discard Decisions

While the variance-aware heuristic described previously provides a simple and effective strategy, it is inherently myopic because it optimizes only the current population state. In contrast, reinforcement learning (RL) can optimize discard decisions with respect to their long-term consequences. The sock system naturally forms a sequential decision-making process. Each discard action influences future cluster composition, age distribution, replenishment events, and variance evolution. Therefore, the problem can be modeled as a Markov Decision Process (MDP).

12.1 MDP Formulation

An MDP is defined by a state space, action space, transition dynamics, and reward function.

State Space A complete state may contain the value and boundary age of every sock:

$$s_t = \{(x_i, k_i)\}_{i=1}^{N_t},$$

where

- x_i is the sock value,
- k_i is the post-boundary age if the sock has already reached 160 or 60.

Since the full state may be high-dimensional, a compact state representation is preferable:

$$s_t = (\mu_W, \sigma_W, \mu_B, \sigma_B, N_W, N_B, A_W, A_B),$$

where

- μ_W, μ_B are cluster means,
- σ_W, σ_B are cluster standard deviations,
- N_W, N_B are cluster sizes,
- A_W, A_B are average boundary ages.

Additional features may include:

- values of the two unused socks,
- ages of the two unused socks,
- number of removals since the last replenishment.

Action Space After selecting and updating the closest pair, two unused socks remain. The agent chooses one of the following actions:

$$\mathcal{A} = \{a_1, a_2, a_3, a_4\},$$

where

$$a_1 : \text{Keep both,} \tag{1}$$

$$a_2 : \text{Discard sock 1 only,} \tag{2}$$

$$a_3 : \text{Discard sock 2 only,} \tag{3}$$

$$a_4 : \text{Discard both.} \tag{4}$$

This small action space enables efficient learning.

12.2 Reward Design

The primary objective is to maintain compact clusters while avoiding excessive turnover. A basic reward function is

$$r_t = -(V_W + V_B),$$

which directly penalizes variance. However, a more realistic objective also considers cluster separation and replacement costs:

$$r_t = -\lambda_1(V_W + V_B) + \lambda_2(\mu_W - \mu_B) - \lambda_3 D_t,$$

where

- V_W and V_B are cluster variances,
- $\mu_W - \mu_B$ measures cluster separation,
- D_t is the number of discarded socks,
- $\lambda_1, \lambda_2, \lambda_3$ are weighting parameters.

This reward encourages:

1. low within-cluster variance,
2. large inter-cluster separation,
3. moderate replacement frequency.

12.3 State Transitions

The environment evolves according to the following sequence:

1. Draw four socks.
2. Select the closest pair.
3. Update the selected pair.
4. Apply the RL discard action.
5. Apply boundary-aging removal.
6. If six socks have been removed, add three White and three Black socks.

The aging mechanism remains

$$P_k = 1 - p^k,$$

where k is the number of selections experienced after reaching the boundary value.

12.4 Expected Learned Policy

A trained RL agent is expected to discover strategies that are difficult to derive manually. For young socks:

$$x_W \approx 255, \quad x_B \approx 0,$$

the agent will typically retain them because they strengthen cluster purity. For mature socks located near their cluster centers, the agent will usually retain them because they stabilize the cluster distribution. For outliers,

$$|x_i - \mu_c| \gg \sigma_c,$$

the agent will frequently discard them because they contribute disproportionately to variance. For socks that have already reached the boundaries 160 or 60, the learned policy will balance two competing options:

1. remove them immediately,
2. allow the natural aging mechanism to remove them later.

The resulting strategy effectively learns when an active discard is worthwhile and when waiting for natural removal is preferable.

12.5 Learning Algorithms

Several RL algorithms are applicable:

- Q-Learning,
- Deep Q Networks (DQN),
- Actor-Critic methods,
- Proximal Policy Optimization (PPO).

Because the action space contains only four actions, DQN is likely sufficient and computationally efficient.

12.6 Research Significance

The resulting framework transforms the sock process from a heuristic clustering problem into a stochastic control problem. The system combines:

- nearest-neighbor selection,
- cluster evolution,
- age-dependent mortality,
- population replenishment,
- reinforcement learning control.

Consequently, the model can be viewed as a controlled birth-death process with online variance regulation. This formulation enables the study of equilibrium variance, optimal replacement strategies, and adaptive population control policies under uncertainty.

13 Conclusion

The final model combines nearest-neighbor selection, variance-aware outlier removal, age-dependent boundary discarding, and periodic population replenishment. The resulting system forms a controlled birth-death process that maintains two compact clusters while continuously eliminating aged and non-representative socks. The aging mechanism

$$P_k = 1 - p^k$$

provides progressively increasing removal pressure and naturally supports long-term variance control.